

Data-Driven Prediction of Traffic Violations Using Improved Random Forest

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Abstract—Traffic violations represent a critical issue for public health and safety globally. This paper introduces a novel machine learning framework designed to predict traffic violations using comprehensive data collected over one year from Qatar. The dataset includes five key features: date, time, zone number, street number, and type of license plate, with the type of traffic violation as the target variable. After data cleaning and preprocessing, including the conversion of input data to numerical values and normalization, the class imbalance is addressed. Various sampling techniques are tested, with random oversampling yielding the best results. This study evaluates several machine learning models, including Naïve Bayes, Logistic Regression, Support Vector Machine (SVM), XGBoost, Neural Network, and Random Forest, for predicting traffic violations. To enhance performance, the proposed method introduces an improved version of the conventional Random Forest by incorporating novel modifications at two critical stages: tree construction and model evaluation. These enhancements aim to increase the accuracy and robustness of predictions, demonstrating the effectiveness of the approach over baseline models. Among these, the improved Random Forest model achieves the highest prediction accuracy at 99.5%. Feature importance analysis reveals that the street number is the most significant predictor, while the type of license plate is the least significant. The experimental results validate the effectiveness of the proposed method, highlighting its potential as a valuable tool for predicting and preventing traffic violations. This study aims to enhance traffic safety measures and demonstrates promise for broader applications in traffic management and safety systems.

Keywords—Machine Learning; Traffic Violation; Data Analysis; Prediction; Random Forest.

I. INTRODUCTION

The safety and efficiency of modern transportation systems have become critical concerns for policymakers and traffic management authorities due to the increasing prevalence of road accidents and traffic violations. As the volume of vehicles increases each day, the number of traffic violations also rises, leading to a surge in traffic accidents and fatalities worldwide [1]. Traffic accidents represent a critical public health issue that necessitates collaboration among various organizations to address effectively [2]. To address these challenges, innovative approaches incorporating predictive analytics and artificial intelligence (AI) have emerged as promising solutions for revolutionizing traffic management and enhancing road safety.

The prediction of traffic violation sparked significant interest within the artificial intelligence community [3-10]. Ding et al. [11] proposed a methodology to forecast accidents within a bus fleet by leveraging violation data. Their study delved into the impact of safety and service violations on vehicular accidents. In a similar vein, Tan et al. [12] presented a model aimed at replicating the influence of driver safety attitudes on instances of traffic violations. This was achieved through the utilization of a questionnaire survey involving drivers, adapted from the modified theory of planned behavior. In a separate study, Joo et al. [13] explored risk assessment for individual drivers, considering both traffic violations and crash records. Their approach involved the training of a Bayesian network to facilitate probabilistic risk assessments. Wan et al. [14] employed a random forest-based approach to tackle traffic violations among taxi drivers, offering insights into significant factors influencing the severity of these violations. The methodology they employed yielded valuable information for controlling the traffic violation. In a different context, Tuna et al. [15] focused on a cellular network for traffic volume prediction, leveraging spatial and temporal data. Their work introduced a novel cost function that factored in peak hours and days of the week, enhancing the accuracy of their analysis. Li et al. [16] used automated enforcement system data to analyze and predict intersection traffic violations, identifying factors such as time of day, traffic conditions, and weather. They found that the Random Forest algorithm excelled in binary prediction (violation happened or not) accuracy and performance with imbalanced data.

In recent years, the application of machine learning, particularly deep learning, in traffic prediction has shown remarkable advancements, enabling more precise and reliable forecasting of traffic conditions. Yin et al. [17] provide a comprehensive survey on various deep learning approaches used in traffic prediction, highlighting the importance of spatio-temporal dependencies and the need for sophisticated models to manage these complexities. Similarly, Kothai et al. [18] introduce a hybrid model combining CNN and boosted LSTM ensemble (BLSTME) to effectively predict traffic congestion by leveraging the strengths of both architectures. Their model achieves high accuracy and performance metrics, outperforming traditional deep learning models. In another significant study, Essien et al. [19] propose a model that integrates traffic data with

social media inputs from Twitter to address non-routine traffic events, improving the accuracy of traffic flow predictions in urban areas by using a deep Bi-directional LSTM stacked autoencoder architecture.

Furthermore, the exploration of intelligent and adaptive systems for traffic management has been a critical focus. Khan et al. [20] present a machine learning-driven, self-adaptive traffic light control system that utilizes image processing via YOLOv3 to manage traffic at signal junctions efficiently. Abdullah et al. [21] emphasize the application of soft GRU-based recurrent neural networks for congestion prediction, which utilize real-time sensor data to classify traffic conditions and enhance prediction accuracy. These approaches are complemented by the work of Xu et al. [22], who propose a multi-view spatiotemporal convolutional network model for traffic flow prediction, showcasing the computational efficiency and strong feature extraction capabilities of convolutional networks.

Additionally, recent studies have explored innovative methods to address the dynamic and fluctuating nature of traffic data. Zheng et al. [23] evaluate hybrid deep learning models in large-scale road networks, offering insights into the architectural designs and performance metrics of these models. Nie et al. [24] propose a Digital Twin framework combining Deep Q-Learning and Generative Adversarial Networks (GAN) for network traffic flow (volume of traffic) prediction, which enhances the prediction accuracy by accounting for time-varying traffic flows. Moreover, Wong et al. [25] review the role of machine learning in mitigating traffic congestion, emphasizing its superiority over traditional methods in optimizing traffic management through route optimization and demand forecasting.

Finally, Liu et al. [26] introduce a federated learning-based intelligent traffic flow forecasting model that integrates spatial-temporal graph-based deep learning within a Multilevel Federated Learning framework. This model addresses privacy concerns by allowing data collaboration without data sharing and demonstrates superior performance in reducing communication overhead and improving prediction accuracy. Collectively, these studies underscore the significant potential of advanced machine learning techniques in revolutionizing traffic prediction and management, thereby contributing to the development of intelligent and sustainable urban transportation systems.

Traditional rule-based systems have some limitations in handling the dynamic of traffic conditions, making machine learning techniques an attractive alternative due to their data-driven and adaptable nature. This research paper introduces a novel method that uses the power of machine learning to forecast traffic violations. It is done by providing valuable insights of traffic management and accident prevention. The main concern of this study lies in the design and implementation of a novel machine learning framework targeted the traffic violation prediction. By integrating multidimensional data with the predictive capability of machine learning, the model can anticipate potential traffic rule breaches and identify high-risk areas prone to violations.

The objectives of this research endeavor are twofold: firstly, to design an efficient and robust machine learning models capable of accurately forecasting traffic violations; secondly, to evaluate the model performance against real-world datasets from diverse geographical locations, traffic conditions, and demographic profiles. The successful implementation of this novel machine learning system is expected to empower traffic authorities with proactive measures to address traffic violations and allocate resources more effectively. Furthermore, this research adds to the growing body of knowledge in the fields of AI, predictive analytics, and transportation engineering, serving as a valuable reference for future studies in related domains. The organization of the paper is as follows: Section II is devoted to the proposed method and the results and discussion are given in Section III. Finally, the paper is concluded in Section IV.

II. PROPOSED METHOD

The effective management of traffic violations is crucial for ensuring road safety and the smooth flow of traffic. Traditional methods of identifying and categorizing violations have their limitations, making the adoption of data mining techniques an attractive proposition. In this section, we present the design and analysis of a machine learning models specifically tailored for traffic violation. In the following, the data analysis, and designing and implementing of the proposed machine learning framework are discussed in detail.

A. Data Analysis

In this section, we delve into the comprehensive process undertaken to analyze traffic violation data from Qatar for the year 2019. Our objective is to transform raw data into a structured format that can be effectively utilized by machine learning algorithms. This involves several critical steps, starting with an initial understanding of the dataset and moving through data cleaning, encoding, normalization, refinement, and balancing. Each of these steps plays a pivotal role in ensuring that the data is of high quality and suitable for building robust predictive models. The following subsections provide a detailed account of each phase in the data analysis process, highlighting the methods and techniques employed to prepare the data for machine learning applications.

1. Data Understanding

The original raw data has been stored in an excel file which encompasses all traffic violations in Qatar throughout the entirety of 2019. Table I illustrates a partial dataset as an example; the full dataset contains approximately 1.2 million records. In this representation, column 1 corresponds to the date, column 2 to the time, column 3 to the zone number (an integer between 1 and 99), column 4 to the street number (an integer between 1 to 3256), column 5 to a binary variable indicating the license plate type, which can be either "Company" or "QID" representing company and government vehicles and finally column 6 to the name of the traffic violation, respectively.

Table I. Sample of Raw Data on Traffic Violations.

Date	Time	Zone No.	Street No.	License Plate Type	Traffic Violation
2019-10-02	9:26:08 PM	74	212	Company	Non-compliance of the vehicle driver with the self-driving traffic lights.
2019-10-02	2:41:04 PM	74	880	QID	Failure to follow the instructions on any sign erected or engraved on the road by the licensing authority.
2019-10-01	10:30:56 AM	92	315	Company	Exceeding the maximum speed limit on the road.
...	...	...	...	...	...

Fig. 1 illustrates the correlation among the dataset variables.

The correlation coefficient ranges from -1 to 1. A coefficient of 1 indicates a perfect positive correlation, meaning that as the value in the first column increases, the value in the corresponding column also increases proportionally. A coefficient of 0.9 signifies a strong positive correlation, where an increase in one value is likely to be associated with an increase in the other value. Conversely, a coefficient of -1 indicates a perfect negative correlation, where an increase in one value results in a proportional decrease in the other. A coefficient of 0.2, however, suggests a weak or negligible correlation, implying that changes in one variable do not reliably predict changes in the other. As can be seen from this figure, the maximum correlation (0.31) exists between zone number and street number, but this correlation is not high enough (at least 0.6) to be considered as a strong correlation.

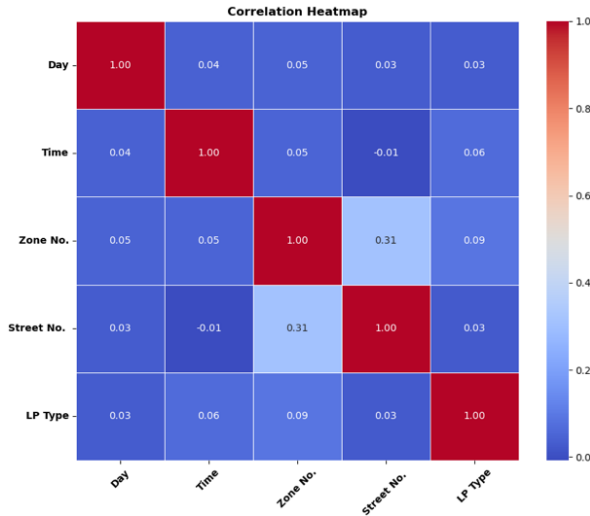


Fig. 1. Correlation between input variables.

2. Data Cleaning

The data in the Excel file contains missing values for the zone and street numbers in some traffic violation records. To ensure the effective training of machine learning models, these incomplete records are excluded from the dataset.

3. Data Encoding

As depicted in Table. I, the data presented in the table is available in diverse formats, which are not directly compatible with the requirements of machine learning models. To address

this issue, a series of mappings are employed to create suitable inputs and outputs for the machine learning training process. The input for the machine learning model consists of the following features: day (an integer number ranging from 1 to 365), time (a float number between 0 and 1, representing the time of day where 0 corresponds to 00:00 and 1 corresponds to 23:59), zone number, street number, and license plate type (1 for QID and 0 for COMPANY). On the other hand, the output is represented by an integer number in range of 1 to 97 that corresponds to each unique traffic violation type. These transformed inputs and outputs enable effective training of the models, ensuring it can make predictions accurately based on the given data. In Tables II, the numeric input and output sets are displayed, showcasing only 20 lines due to the page limitation.

Table II. Inputs and outputs of machine learning models (only 20 elements shown).

Input					Output
Date	Time	Zoon No.	Street No.	License Plate Type	Traffic Violation
359	0.7361	96	340	0	67
291	0.6528	80	800	1	34
255	0.3333	94	500	0	64
272	0.7083	80	800	1	39
272	0.7403	80	800	0	39
309	0.4583	86	100	1	33
254	0.4784	80	800	1	39
277	0.4479	80	800	1	72
276	0.5472	80	800	1	39
258	0.7917	90	760	1	39
364	0.8327	66	51	1	45
362	0.3470	53	530	1	45
362	0.7212	53	530	1	45
361	0.2133	74	212	1	45
362	0.4592	74	212	1	45
362	0.7719	74	212	1	45
362	0.9569	74	212	1	45
361	0.6857	74	212	1	45
361	0.9465	74	212	1	45
363	0.6707	82	1800	1	45
...	...	...	...	...	...

4. Data Normalization

To ensure that all features contribute equally to the analysis and to improve the performance of machine learning algorithms, we normalize the dataset. Each numerical value in the dataset is scaled to a range between 0 and 1. This transformation is performed using min-max normalization, which rescales the data based on the formula:

$$x_n = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

where x represents the original value, x_{min} and x_{max} are the minimum and maximum values of the feature, respectively, and x_n is the normalized value. This normalization process ensures that the features are on a comparable scale, which is crucial for many machine learning algorithms.

5. Data Refinement: Filtering Rare Traffic Violations

Fig. 2 presents a histogram of traffic violations. It is evident from the figure that some traffic violations occur much more frequently than others, while some are quite rare. To refine the dataset, traffic violations occurring fewer than 10 times per day across the entire country are excluded. This filtering step resulted in the removal of only 0.44 percent of the total records.

Consequently, nine traffic violations, accounting for 99.54 percent of all records, are retained. These traffic violations are listed in Table III.

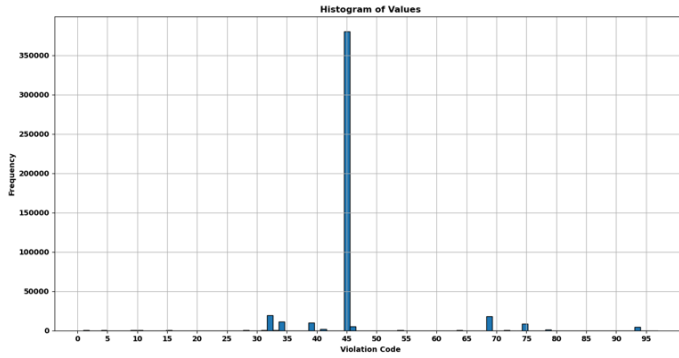


Fig. 2. Histogram of traffic violations from Table II.

6. Dataset Balancing

As can be seen in Table III, traffic violation code 45 occurs 380,183 times, while traffic violation code 41 occurs only 1,670 times. Therefore, our dataset is unbalanced. Balancing an imbalanced dataset is crucial for training machine learning models effectively, as imbalanced datasets can lead to models that are biased towards the majority class. For balancing the dataset, techniques based on oversampling the minority class, under sampling the majority class, and a combination of both are tested in this paper.

Table III. Most frequent traffic violations occurring more than 10 times per month.

Traffic Violation	Traffic Violation Code	Frequency
Exceeding the maximum speed limit on the road.	45	380183
Parking or waiting for vehicles in places designated for pedestrian crossing or on sidewalks.	32	19621
Failure to follow the instructions shown on any sign erected or engraved on the road by the licensing authority.	69	18042
Parking or waiting of vehicles in places where parking and waiting are not permitted.	34	11380
Parking and waiting for mechanical vehicles in places other than those designated for that.	39	9724
The vehicle driver's failure to adhere to self-propelled traffic lights.	75	8257
The vehicle driver moves his vehicle and confuses or disrupts traffic at intersections with traffic signals.	46	4971
The vehicle driver's failure to take into account the following: leave a sufficient distance between himself and the vehicle in front of him and stand It follows its driver's signals, and he is not permitted to pass it except to the left.	94	4235
The vehicle driver left his vehicle on the road without taking the necessary measures to avoid any accident or leaving it in a way that would lead to it. To obstruct traffic or leave it without taking the necessary measures to make illegal use by others impossible, or He left it with the switch on it.	41	1670

Under sampling reduces the size of the majority class by randomly removing instances, which can lead to faster training times and lower computational costs. It effectively addresses class imbalance and reduces the risk of overfitting since the dataset size is smaller. However, this method can result in a significant loss of valuable information from the majority class, potentially making the remaining data unrepresentative and leading to underfitting. The reduction in data can also negatively impact the model overall performance if critical patterns and variations within the majority class are lost.

Oversampling increases the number of minority class instances by duplicating existing instances or generating synthetic ones, such as through SMOTE (Synthetic Minority Over-sampling Technique) [27]. This helps ensure the model has enough examples to learn from the minority class, improving its performance and generalization on these instances. While retaining all majority class information, oversampling can lead to longer training times and higher computational costs due to the increased dataset size. Additionally, there is a risk of overfitting. We test using Random oversampling and SMOTE for oversampling, and random under sampling for under sampling. Based on our experiments (discussed in the next section), the validation accuracy of the machine learning algorithms significantly improves with the use of random oversampling techniques. This improvement is due to under sampling leading to the loss of important information in our dataset, while oversampling techniques retain it. Figure 3 shows the histogram of the balanced dataset.

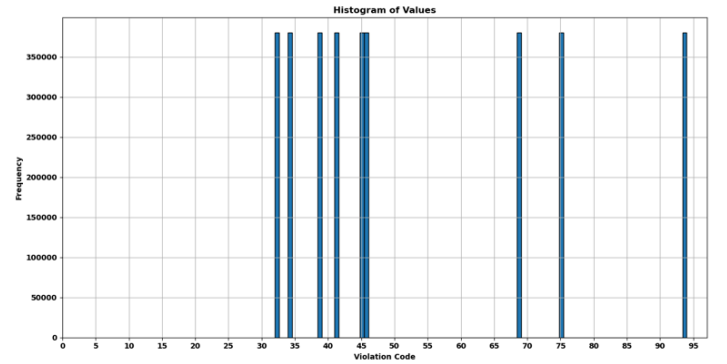


Fig. 3. Histogram of balanced dataset using the random oversampling method.

B. Machine Learning Models

In this section, we present the machine learning models utilized for predicting traffic violations to identify the model that best fits our dataset. The implemented models are Naïve Bayes [28], Support Vector Machine (SVM) [29], Logistic Regression [30], XGBoost [31], Random Forest [32], Neural Network [33], and the improved Random Forest.

1. Naïve Bayes

Naïve Bayes classifiers are based on applying Bayes theorem with strong (naïve) independence assumptions between the features.

2. Support Vector Machine (SVM)

Support Vector Machine is a powerful supervised learning model that can be used for both classification and regression challenges. It works by finding the hyperplane that best divides a dataset into classes.

3. Logistic Regression

Logistic Regression is a statistical model that in its basic form uses a logistic function to model a binary dependent variable. It is simple and efficient for binary and linear

classification tasks. In our case, we extend it to multi-class classification.

4. XGBoost

XGBoost (Extreme Gradient Boosting) is an optimized distributed gradient boosting library designed to be highly efficient, flexible, and portable.

5. Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the mode of the classes for classification. It is known for its robustness and high accuracy.

6. Neural Network

Neural Networks are a set of algorithms modelled loosely after the human brain, designed to recognize patterns. They interpret sensory data through a kind of machine perception, labelling, or clustering of raw input. Before training the neural network, the class labels are transformed into a categorical format using label encoding and one-hot encoding techniques.

7. Improved Random Forest

Conventional Random Forest algorithms typically employ the CART method to generate decision trees and determine the splitting criteria. While they use the Gini index, which is computationally efficient, it may not adequately capture key relationships between features relationships that are essential for prediction tasks involving complex and interacting factors. In this paper, Gini is replaced with Information Gain (*IG*), a criterion that quantifies the reduction in entropy resulting from a feature split. It is defined as:

$$IG(D, A) = H(D) - H(D, A) \quad (2)$$

where D is the dataset, A is the feature used for the split, $H(D)$ is the entropy of the dataset before split, and $H(D, A)$ is the weighted entropy of the dataset after splitting on feature A . For a dataset D with n distinct classes, the entropy is given by:

$$H(D) = - \sum_{i=1}^n p_i \log_2(p_i) \quad (3)$$

By relying on Information Gain, the improved Random Forest selects features that provide the highest information gain, thereby better reflecting important underlying patterns in the dataset. This is critical in traffic violation prediction, where subtle feature interactions can influence the outcome. The use of Information Gain aligns with Shannon's information theory, which emphasizes that reducing uncertainty improves decision-making. Traffic violations often involve complex, unpredictable behaviors, and a more expressive split criterion like Information Gain enables the model to capture these behaviors more effectively than Gini.

Another modification is incorporating the Out-of-Bag (OOB) error estimation mechanism which introduces a built-in mechanism for unbiased model evaluation during training. Each tree in the forest is trained on a bootstrapped sample D_k drawn from the full dataset $D = \{(x_1, y_1), \dots, (x_N, y_N)\}$, with approximately 63.2% of the samples used, leaving around

36.8% of the data forming the OOB set for that tree. The predictions on these OOB samples are aggregated across the forest to provide an unbiased estimate of the generalization error. For any sample x_i , the OOB prediction is obtained by majority voting:

$$\hat{y}_{oob,i} = \arg \max_{c \in C} \sum_{T_k \in \tau_i} I(T_k(x_i) = c) \quad (4)$$

where $\tau_i = \{T_k: x_i \notin D_k\}$ is the set of trees where x_i is in the OOB set and C is the set of possible classes, and I is the indicator function:

$$I(condition) = \begin{cases} 1 & \text{condition} = \text{True} \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

The overall OOB error is then calculated as:

$$\text{OOB Error} = \frac{1}{N} \sum_{i=1}^N I(\hat{y}_{oob,i} \neq y_i) \quad (6)$$

This OOB mechanism ensures that the model continuously evaluates its performance on unseen data, eliminating the need for a separate validation set. It also helps capture rare events such as specific types of traffic violations by ensuring that even less common samples contribute to the model evaluation.

In combination with OOB error estimation, the use of Information Gain further enhances the model's performance. Information gain prioritizes feature splits that reduce uncertainty the most, enabling the improved Random Forest to better identify patterns relevant to traffic violations, such as behavioral trends or situational conditions. This ensures that the model generalizes well across both frequent and rare events, offering more accurate and meaningful predictions. The improved Random Forest integrates more informative feature selection with adaptive error estimation, making it a robust and practical tool for traffic violation prediction. By refining both the splitting criterion and evaluation process, the model achieves higher accuracy and better generalization to unseen cases, ensuring reliable performance in real-world scenarios.

Fig. 4 illustrates the flowchart of the proposed method. As can be seen, the proposed method for predicting traffic violations follows a structured pipeline to enhance model performance. Initially, the dataset undergoes data encoding and normalization to ensure consistent input formats for machine learning. Infrequent violations are then refined to mitigate data sparsity issues, followed by oversampling to address class imbalance. The model leverages bootstrap sampling to build multiple decision trees, calculating information gain to select the most relevant features.

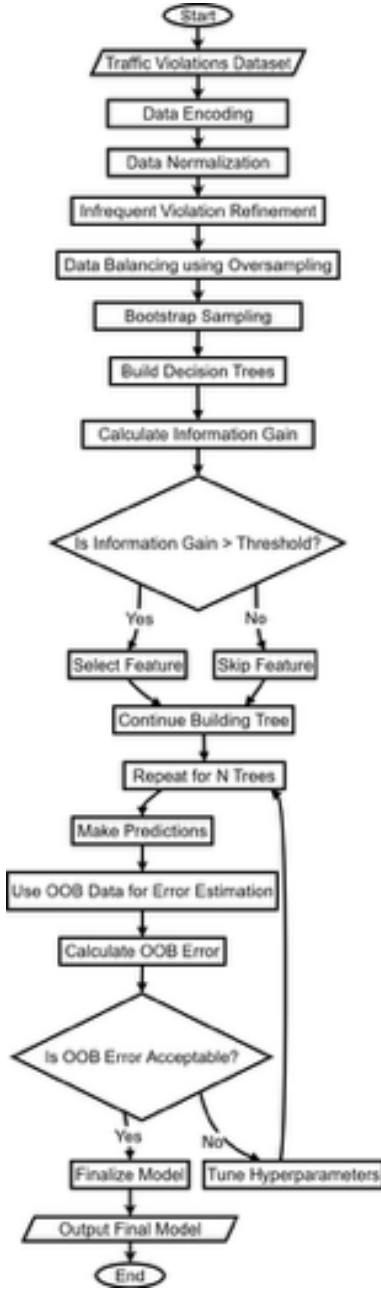


Fig. 4. Flowchart of the proposed method.

If the information gain exceeds a predefined threshold, the feature is included; otherwise, it is skipped. This process is repeated across N trees for robust training. Predictions are validated using Out-of-Bag (OOB) data, where the OOB error is estimated. If the error is acceptable, the model is finalized; otherwise, hyperparameter tuning is performed iteratively until the desired accuracy is achieved. The final model is then output as the end product of this process.

III. EXPERIMENTAL RESULTS

A. Model Training

In this section the machine learning models are trained over the balanced traffic violation dataset. 80 percent of data are used for training and rest 20 percent is used for validation purposes. Based on a vast number of experiments, the following parameters lead to the best performance for the machine learning models we test. For Naïve Bayes, despite its simplifications, Multinomial Naive Bayes performs surprisingly well in various complex real-world situations. For SVM, we use a radial basis function kernel with a regularization parameter (C) of 1.0 and enabled probability estimation. In Logistic Regression, we set the maximum number of iterations to 1000. For XGBoost, we use 100 estimators, a learning rate of 0.1, disabled label encoding, and evaluate performance using the multi-class log loss metric. The number of estimators (trees) is set to 100 for both conventional and improved Random Forest methods. The architecture of our neural network includes an input layer with 2048 units and ReLU activation, three hidden layers with 512, 128, and 32 units respectively (each with ReLU activation), and an output layer with 9 units (corresponding to the number of classes) and SoftMax activation. We use the Adam optimizer with a learning rate of 0.001, the categorical cross entropy loss function, accuracy as the evaluation metric, and train the model for 100 epochs.

Table IV shows the effect of different balancing methods on the accuracy of various machine learning models. As can be seen in this table, random oversampling is the best balancing method for the top competitors.

Table IV. Effect of balancing methods on the accuracy of machine learning models.

Balancing Method	Naïve Bayes	SVM	Logistic Regression	XGBoost	Neural Network	Random Forest	Improved Random Forest
Random Under sampling	25.88	32.30	28.38	72.19	65.37	70.53	71.82
SMOTE Oversampling	23.51	33.54	32.01	74.96	69.34	75.44	76.35
Random Oversampling	25.19	33.98	31.99	76.69	80.85	85.16	99.50

Table V presents the quantitative comparison of the machine learning models, with results derived from cross-validation to ensure robustness and reliability. The key metrics used in this Table: accuracy, precision, recall, and F1-score, are defined as follows.

Table V. Quantitative comparison of various machine learning models.

Measurement	Naïve Bayes	SVM	Logistic Regression	XGBoost	Neural Network	Random Forest	Improved Random Forest
Accuracy	25.19	33.98	31.99	76.69	80.85	85.16	99.50
Precision	21	22	23	76	81	0.85	100
Recall	25	34	32	77	81	0.85	99
F1-Score	19	26	25	76	81	0.85	99

Accuracy is the ratio of correctly predicted observations to the total observations. It is a measure of the overall effectiveness of a model.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (7)$$

where TP , TN , FP , and FN indicate true positive, true negative, false positive and false negative, respectively.

Precision, also known as positive predictive value, is the ratio of correctly predicted positive observations to the total

predicted positives. It indicates the accuracy of the positive predictions.

$$Precision = \frac{TP}{TP+FP} \quad (8)$$

Recall, also known as sensitivity or true positive rate, is the ratio of correctly predicted positive observations to all observations in the actual class. It measures the ability of the model to capture positive instances.

$$Recall = \frac{TP}{TP+FN} \quad (9)$$

The F1-Score is the harmonic mean of precision and recall. It is a useful metric when the class distribution is imbalanced. The F1-Score balances the trade-off between precision and recall.

$$F1 - Score = 2 \frac{Precision \times Recall}{Precision + Recall} \quad (10)$$

As Table V confirms, with an accuracy of 99.50%, a perfect precision of 100, and high recall and F1-scores of 99, the improved Random Forest model proves to be the most effective for the given task. In contrast, models like Naïve Bayes and Logistic Regression exhibit poor performance across these metrics, while conventional Random Forest, XGBoost and Neural Networks show notable improvements but still fall short compared to improved Random Forest.

Given these results, the improved Random Forest model is clearly the optimal choice for further experimentation, offering near-perfect prediction capabilities and ensuring minimal false positives and missed true positives. Its superior performance justifies its selection as the primary model for the subsequent phases of the experiment.

In this study, we employ the Receiver Operating Characteristic (ROC) curve to further evaluate the performance of multi-class classification models. Fig. 5 shows the ROC curve for the models presented in Table V. The ROC curve is a graphical representation that illustrates the diagnostic ability of a classifier by plotting the true positive rate (sensitivity) against the false positive rate (1-specificity) at various threshold settings. For multi-class classification, we utilize the one-vs-rest (OvR) approach, which involves generating an ROC curve for each class against all other classes. This method enables a comprehensive evaluation of the classifier performance across all classes. Each ROC curve is plotted, and the area under the curve (AUC) is calculated to quantify the classifier performance for each specific class. A higher AUC value indicates a better model performance, with a value of 1 representing perfect classification. These ROC curves and corresponding AUC values provide valuable insights into the strengths and potential areas for improvement in our classification models. The results demonstrate that improved Random Forest model achieve high AUC values across all classes, indicating robust predictive capabilities.

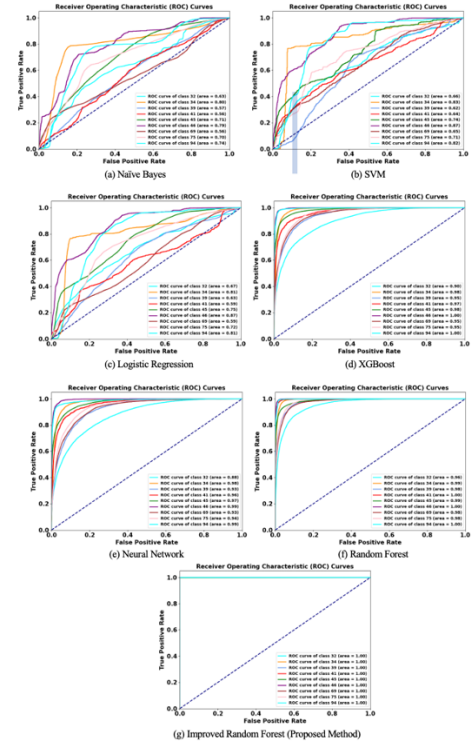


Fig. 5. Receiver Operating Characteristic (ROC) curves of various machine learning models.

Figures 6 and 7 provide a detailed evaluation of the improved Random Forest model performance in predicting traffic violations. Figure 6, the confusion matrix, reveals the model high accuracy, with most values on the diagonal being 1.00, indicating perfect classification for those traffic violation codes. Misclassifications are minimal and infrequent, as evidenced by the low off-diagonal values, highlighting the model robustness and precision in distinguishing between different types of traffic violations.

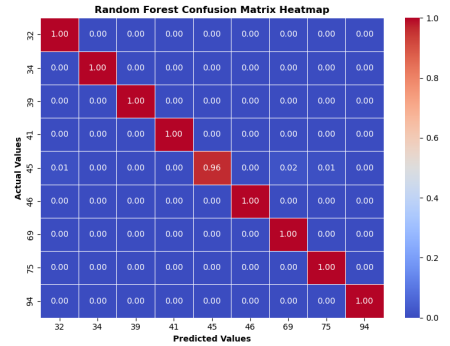


Fig. 6. Confusion matrix of the improved Random Forest method. The labels are traffic violation codes.

Figure 7, the feature importance diagram, ranks the features based on their contribution to the model predictions. Street No. emerges as the most significant feature, followed by Time, suggesting that the location and time of day are crucial in predicting traffic violations. Zone No. and Day also play important roles but to a lesser extent, while LP Type is the least influential. This hierarchy of feature importance underscores the necessity of focusing on key factors like location and time to enhance traffic management strategies and improve prediction accuracy. Together, these figures demonstrate the efficacy of the improved Random Forest model and provide insights into the pivotal factors influencing traffic violations.

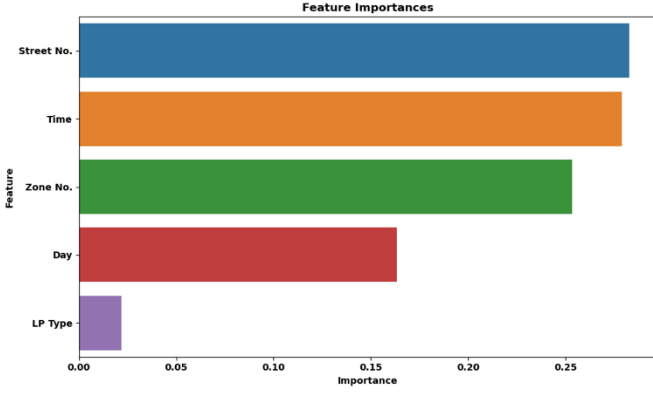


Fig. 7. Feature Importance diagram.

B. Traffic violation prediction

Using the improved Random Forest technique introduced in the previous section, traffic violations can successfully be predicted. For instance, consider a specific scenario where the question is about traffic violation code 45, which is ‘Exceeding the maximum speed limit on the road’. We want to determine the zone with the highest number of violations for this particular code on September 1, 2023, at 17:00. Figure 8 represents the histogram distribution of traffic violations for code 45 at different zones in Qatar. It provides insights into the areas where the mentioned traffic violation is most prevalent. Additionally, to enhance the visual representation, the distribution of traffic violations can also be displayed on the map of Qatar. This geographical representation offers a clearer understanding of the regions with the highest occurrences of the specified violation code. Using the information about zones of Qatar along with their respective latitudes and longitudes, and considering Figure 8, a heat map is produced to illustrate the distribution of traffic violations on the map of Qatar. The resulting visualization will be displayed as shown in Figure 9. The heat map uses color gradients to represent the intensity or density of traffic violations in different zones of Qatar. Hotter colors like red and orange indicate higher concentrations of violations, while cooler colors like blue and green represent lower densities. This graphical representation provides a clear and intuitive view of the areas where traffic violations are more prevalent, helping to identify zones with potential traffic management issues and aiding in decision-making for targeted interventions or improvements in road safety and traffic flow.

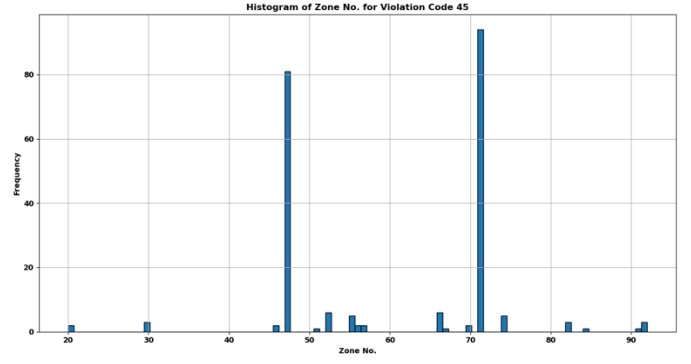


Fig. 8. Histogram of traffic violation code 45 on September 1, 2023, at 17:00.

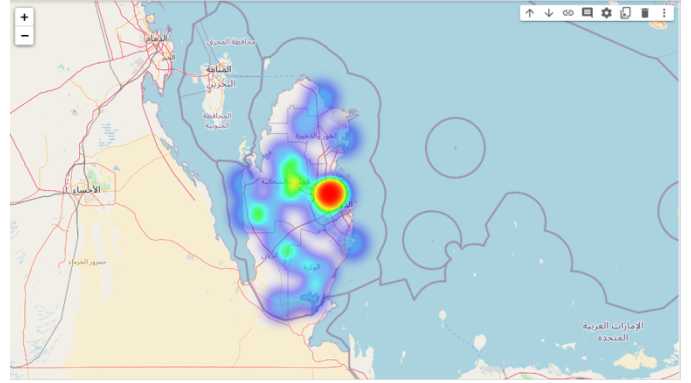


Fig. 9. Heat map of traffic violation code 45 on September 1, 2023, at 17:00.

IV. CONCLUSION

This research presented a comprehensive machine learning framework aimed at predicting traffic violations, using an extensive dataset collected over one year from Qatar. By focusing on five critical features—date, time, zone number, street number, and type of license plate and targeting the type of traffic violation, the research demonstrated the feasibility and effectiveness of predictive modeling in the realm of traffic safety. Through meticulous data cleaning and preprocessing, the dataset was refined to enhance model performance. The reduction of traffic violation classes from 97 to 9, while maintaining 99.6% of the records, ensured a focused and manageable prediction task. Addressing class imbalance through various sampling techniques, with random oversampling proving to be the most effective, further contributed to the robustness of the predictive models. Among the evaluated machine learning models, the improved Random Forest classifier emerged as the most accurate, achieving an impressive 99.5% prediction accuracy. This highlighted the potential of the proposed framework in accurately identifying traffic violation patterns. The feature importance analysis underscored the critical role of the street number in predicting violations, while the type of license plate was deemed the least significant. The experimental results validated the proposed method, showcasing its potential as a valuable tool for predicting and preventing traffic violations. By leveraging these

predictive insights, traffic management authorities could implement more targeted and effective safety measures, ultimately enhancing public health and safety.

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